

ActInf GuestStream #001.1 ~ Majid D. Beni "A critical analysis of Markovian

GuestStream | Majid D. Beni | 1:12:41

so uh hello i'm

maggie oh we all all started

all right so hello

hi everyone welcome to the active

inference lab

and to the first active inference guest

stream

i'm here with our special guest majeed

benny

and we're just really honored and

excited to be hearing your presentation

we're going to be having a about 30

minute presentation

on whatever you'd like to be speaking on

and then i'm going to be collecting

questions from the live chat and then

relaying them to you

and it was already just fun to talk to

you right before we started

so thanks again majeed and take it away

go for it

all right uh hello thank you very much

uh

yeah i'm very excited too and uh i thank

you very much for giving me the chance

to present a part of my work so

uh i'm basically beginning to speak uh

what i've done um a part of my work

uh on a free energy principle so

i will try to share my screen right now

because i'm a great believer in

extended mind and i barely could do

anything without

having my screen shared so you you

mostly able to see my

screen though yep it looks perfect thank

you perfect thank you

so uh uh

i'm basically going to present uh a part

of my work in a recent paper in centers

critical analysis of mercury and monism

and the structure of this talk is quite

uh a straightforward

so uh i'm not uh i briefly

uh very briefly and with a very broad

branch we'll speak about free energy

principle and mercury and monism

so i gloss over the technical stuff

because

for one thing i'm a philosopher and i'm

not very well versed in these technical

aspects and for everything i'm going to

present the core of my work which is

about scientific models

uh so uh after that i speak briefly

about scientific models and

scientific modeling in philosophy of

science and coming from the

background of philosophy of science is

what uh

but it is the main part of my

contribution to this literature

linearity principle to

look at it through the lens of

philosophy of science and issue of

scientific representation

um then i'm going to um

describe markovian models as some sort

of scientific models and

scrutinize their relationship with their

target model in the

real world and see whether any of these

descriptions of the relationship between

mercury and models and their target
system could support smart coffee and
monism
and briefly at the end i will also
evaluate
friston and colleagues attempted
developing markovian moonism along the
lines of structural realism
so um
so most probably you know more than a
free energy principle
and predictive processing uh the idea
has been reintroduced to
uh science but so the roots are
this idea seems to be quite old but uh
it it has been or introduced to resort
to
science by car freestyle and colleagues
and it is all based on
simple and subtle idea so
uh generally
it provides a framework that explains
how a biological system exposed to
random and
unpredictable fluctuation in its
external environment can restrict itself
to occupying
a limited number of a state and
therefore
therefore survive in some recognizable
form so
uh in order to uh be able to resist the
uh dispersing uh force of the
environment the organism have to
in a nutshell have to keep the amount of
its free energy
low and uh predictive processing is some
sort of

theory of application indicating that
when applied to the brain of
some kind of system such as ourselves in
all
free energy minimizing free energy
uh could be specified in terms of uh
minimization of prediction error so
there are these generative models in the
brain and
through a process of a top-down process
the brain
or the organism endeavor to uh
apply these models to reality or at
least to check them against real against
the
evidence that it receives from the
external world
um and then there is discrepancy it
endeavors to
uh uh
somehow revise the models update its
model or at least make changes to the
environment so there is
uh some sort of action by the covers
mental action or over attacking but
active influence is the uh other
side of minimization of prediction error
and uh princeton and colleagues mainly
prison
um developed free energy principle into
a fascinating theory of consciousness in
terms of
uh temporary richness or temporal depth
of
generative model so not only some
organism can
minimize their free energy and
optimize the representation of the

environment they could also
somehow form some sort of a foresight or
prediction about the possible outcomes
of their action in the environment and
it seems that this capacity is somehow
tied
up with conscious properties or
consciousness
and of course there is this idea of uh
mark of blankets
uh judah appeal introduced them i think
in
1980s and there are some sort of
bayesian networks uh which could help us
to make a distinction between
conditional independent account for the
conditional independence between
uh internal and external spaces
so uh the literature itself
uh free energy principle has been quite
flourishing
uh quite a number of scientists and
philosophers
uh have uh contributed to developing
this literature
and amongst the philosophers up to just
to mention
a few of them are great people who
contributed such as
uh jakub pauling andy clark sean
gallagher and
uh even people who are coming from a
background of philosophy of science such
as stefan hartman
have contributed to this uh rich
literature
uh but uh what what's interesting is
that the

free energy critical has not been
developed into some sort of
metaphysical test is about cognition or
consciousness or brain
so as far as i can see
this uh paper
which is the target paper of my paper
incentives from cartesian duality to
markovian modernism is the subtitle
of prison and whis and hobson uh as far
as i can see
it is uh the main work the main paper
uh that actually contributed to
uh spelling out the metaphysical
uh implications of the free energy
principle this is the main paper as far
as i can see that
try to develop free energy principle
into some sort of
metaphysical uh testis of consciousness
and i think that for this reason alone
the paper
deserves uh attention and uh i
i'm a great admirer of the sport i
usually do not engage with
uh reviewing or criticizing single
papers but
this paper attracted my attention and
interest because i think that
uh this is so even before this uh people
such as for
for example fristen and thompson
themselves try to develop a free energy
principle into a theory of consciousness
that provides a point to
um cartesian dualism uh
try to do so but as far as i can see
here for the first time free energy

principle has been developed into
a very uh well-defined
metaphysical thesis which is monism
so the main point here is that i'm more
sympathetic
to the goal of this paper to the
objective of this paper that at times
the paper may
indicate and what i'm going to say
uh is mainly uh aim to
invoke provoke the free energy
principles to develop this markovian
monism into
a slightly a
better uh account a well-supported
account of consciousness metaphysical
account of consciousness
and i'm going to look at this issue
through the lens of philosophy of
science to see the relation between
markovian
blankets and other environments as some
sort of relation between scientific
models because markovian models are some
sort of scientific models and i'm going
to
uh argue for provide some argumentations
for that
uh to draw a connection between
uh markovian models on the one hand and
the target system on the other hand and
so-called consciousness that appears
through this relationship
uh could be uh put into some sort of
modernism which is a metaphysical test
so i'm going to look at this issue
through the lens of philosophy of
science

and unfold the implication
as part of the implication for markovian
monism so
the test is itself markovian monism uh
is quite a straightforward i have
highlighted the part of it for you here
so the thesis is is that fundamentally
there is only one type of thing
and only one type of irreversible
property
and markovian blankets form the calculus
of belief which is not
only a nice scientific mathematical
model of consciousness but also provides
a clear
metaphysical statement and i can shed
light on
current philosophical debates so um the
highlighted parts
so it is this is the statement of
markovian monism
and the red part indicates that
oh how forcefully the
theory have been uh spelled out in
metaphysical terms so
there is only one type of thing is a
strong metaphysical
thesis and this is something that has
been also stated
in a frieza and colleague's paper
so and i here i highlighted another part
of the same phrase
which indicates that
this statement of consciousness on the
basis of markovian models is not just a
nice scientific
or mathematical model of consciousness
but also provides a

metaphysical statement so the main question here is that how could uh hong kong that we could build hong kong that we could read of a metaphysical thesis on the basis of or scientific models of the phenomena scientific mathematical models of phenomena uh so i'm going to uh um ask this question as i say through the lens of philosophy of science to see what's the relation between i i think that this basic question of philosophy of science what's the relation between scientific models and their target system they are significantly on the claim of markovian monism which tries to develop some sort of a strong metaphysical thesis on the basis of this efficient modeling tool scientific modeling tool which are mark of blankets so this is an insight from a philosophy of science and the insight has been i think that originally it has its roots in works of uh ronald geary and uh uh paul churchland for example uh and a number of and bosnian foreign himself a number of important philosophers of science who have been developing this idea in a model-based approach to scientific

tourism

uh 1980s and it has been recently

uh surfaced

uh resurfaced in works of michael

weisberg

and uh roman frigg and

uh god frieza so and the general idea is

that scientific practice is basically

consists of model making making

scientific models

and the thing is that so if you have

when engaging in scientific practice we

just do not directly engage in

in reading of the ideas from the feature

of word

uh reading of the theories the theories

do not just

latch on to the word directly there is

some sort of representation

and the representational relation

between scientific theory scientific

models

and the world is in indirect so

scientific theories do not represent the

features of the work directly

the main thing here is that uh quite a

few

uh human factors interfere through the

process of scientific representation

so to name only a few for example

scientist

interests and explanatory uh goals and

intentions uh play a role in how

scientists are constructing their

scientific models and how use them to

represent the features of reality by

scientific models and as

uh well free for example among other

people explain this
you know process of indirect
representation
is quite sophisticated it includes
several
stages such as denotation or designation
example
exemplification kingdom and imputation
so
uh the main point here the take-home
point here is that the relationship
between scientific models on that one
and
our reality on the other hand or the
target system on the other hand is quite
uh sophisticated it is never
straightforward
and there are three ways of so and mark
mark of blankets markovian models are
just
efficient scientific tools in the
modelers of
toolbox that allow the modeler to
represent the features of the world in
indirectly so i'm going to speak about
three ways of
uh understanding markovian models um
i'm beginning so uh as the sketch that i
showed earlier
uh in the case i'm beginning with uh
interpreting markovian models as
physical or material models first and as
fictional models and
finally as mathematical models and this
order has been chosen deliberately
because i'm going from
the at least likely reading of markovian
models i'm going to are the most likely

interpretation of markovian models
so i'm convinced that markovian models
could be understood this
as some sort of mathematical models
but i also consider other options
because there are some sort of
textual evidence to support them so for
example we could it is indeed possible
to understand more
blankets or markovian models as some
sort of
physical models material models for
example christian says in uh
uh 2019 paper that
the marco blanket is not some aesthetic
statistical device by which we could
or we use it to observe or model the
world it is the
necessary attributes of the universe
that
can be carved into things
so at least to a first approximation we
try to read this phrase or the second
phrase from the same paper it seems that
uh for uh free energy principle theories
for freedom for example
a mark of blankets are actually features
of the reality themselves they are
physical things there are real
real things out there and
so uh i think that it is not uh only
freezing is
advocating this uh reading i think that
also
i i have seen similar phrases in for
example
these and medsing's introduction to
predictive processing and also in

allen and freestone's papers since uh
2060 i think so uh this is
there is some textual basis for
understanding markovia but marco
blankets as physical models
but uh i think
and i think that this indication this
reference to mark of blankets as
physical things has to be taken with
a grain of salt i do not think that
there are actually such things so if we
for example
open the brain or open an organic system
i do not think that we can find anything
resembling a markovian blanket there
actually
there is no blanket there there is no
such physical thing
but i can't understand if a terrorist
uh could conceive of
the target system as if it really
include this
uh representational device but the point
is that even if we consider somehow
considered that markov blankets
markovian models are
uh features of reality or physical
things that exist in the world
even if we can't say that it doesn't
follow that
reality itself shares the ontological
and material features of markov balance
it doesn't follow that uh
there is the same uh
cells same ontological category that
includes both markov blankets and their
target system and there is an example
that could demonstrate that so

for example there is a such thing as a
hydraulic pump
model of economics so hydraulic pump is
a physical object
and for hydraulic model to work some
properties of economic system mostly
identified with
uh hitlering properties of the pond it
could be for example
uh exemplified through the process of
imputation we could
ascribe or attribute some of the
properties of the literary poem to the
physical object
uh sorry
to to ascribe some of the properties of
israeli poem to the economical system
that is supposed to be represented
but it doesn't follow that the
economical system and hydraulic pump
which is the physical objects
uh can be put under the same ontological
category or material category so for
example hydraulic foam which is a
physical object
is made of for example steel or plastic
or some
some other material thing it doesn't
follow that economy shares this
material antique futures
uh so uh i i simply uh put the
uh materialist interpretation
of mark mark of blankets aside
and proceed to consider markov blankets
as fictional models
so and the main reason for this is that
i assume that if we
uh for example explore the actual brain

of organism we couldn't find anything
resembling a mark of blankets inside the
brains
so we could interpret them as some some
sort of
fictional objects and
uh so and
the fictionalist and again there are
textual support for this for example
christian
and colleagues say that if if we
look into the behavior of creatures
through the lenses of marconi and models
it will look as if it is acting the
creature
is acting to minimize its particle or
surf right zone so
this as if phrase luke has a phrase uh
i think that is very important it
indicates that they do not really think
that there is such thing as marco
polanki's in the creature but
uh it's look as if there is such thing
and
um it has been this interpretation this
fictionalist
instrumentalist interpretation
is gaining momentum as far as i can see
so there are a number of
other people such as maxwell ramstad and
uh venus uh that are developing some
sort of fictionalist and instrumentalist
interpretation
of free energy principle but the thing
is that if you concept this fictionalist
interpretation of
marco blanket our free energy principle
uh it follows that so if marco blankets

are used for fictions we couldn't uh
derive any sort of metaphysical
conclusion from them so
from a fiction so for example if you are
studying uh greek mythology
and assuming that greek mythology is
some sort of fiction
you cannot derive any conclusion about
the nature of deity
from what you read in a story from greek
mythology
there are no metaphysical conclusions to
be derived from
fictions and stories and this is
precisely the point
in invoking a fictionalized approach to
scientific models so
people are using fiction assuming that
scientific models are fiction
are useful instruments to just get rid
of the metaphysical questions about the
nature of fiction
models or their relationship with
reality so
to make a longer story show this
fictionalist interpretation doesn't lead
to any sort of metaphysical
conclusion about the nature of
consciousness or the nature of mind or
anything of that sort
so this brings us to
the last interpretation of markov
blankets and it comes in terms of
doesn't see how much time or everything
i don't have much time i try to
wrap it up quickly so uh the the final
interpretation
is uh to interpret

markov blankets in terms of mathematical models

uh and i think that there is good textual growth so uh

it is to to at least to my mind to some extent from for someone who is coming from a background or philosophy of science

it seems rather obvious that marco collected are just mathematical modeling tools so they are

mathematical tools in the context of bayesian networks to describe to represent

their conditional uh

independence between internal and external

spaces uh and there is uh

textual evidence for that again i have highlighted part of that

which indicates that their enterprise aims to describe a construct from information theory

and mathematical physics

that helps them to define

the properties of living system and their cognition

and give them some sort of meaningful interpretation by invoking

uh mathematical objects that could be used to

consolidate this interpretation so

uh and the use of so

i i think that there is some sort of enlightenment

inside uh in modern philosophy of science

uh in the wake of works of patrick soups

and basman for us and in 1960s and 1970s
uh so people's attention has been drawn
to
the importance of mathematical models
sec theory and model theory and that
sort of stuff
uh for modeling natural phenomena
so when doing science scientists
so the rise of modern in some sense the
rise of modern science
uh has been a result of invoking these
sorts of mathematical approaches that
allow us to
represent and model natural phenomena of
you know
precision and vigor and rigor but
uh the thing is that if we assume that
markovian models are some sort of
mathematical
tools that can be used to represent the
features of reality
the thing is that we couldn't develop
any sort of metaphysical conclusion
about the nature of reality on the basis
of application of these mathematical
tools
look if we assume that markov blankets
are
mathematical tools and if we want to
identify their nature with the nature of
reality the only conclusion that follows
is that the reality is some sort of
mathematical objects it's some sort of
mathematical entity
and interestingly enough as i will argue
this is precisely the same sort of
conclusion that
uh that uh fristen and colleagues

uh reached uh
in the end of their paper so uh at some
sense they are saying that they are
in they intend to defend some sort of a
reductive
materialist or redacted physicalist but
what happens is that
they come to the conclusion that because
these two ways of describing the nature
of reality this division on the basis of
markovian monuments
to external and internal spaces because
uh none of these
interpretations known of these
geometries that they develop none of
this
intrinsic and extrinsic perspective
could be
supported under the assumption that
neither perspective
is privileged one would have to conclude
that reality is fundamentally
ontologically neutral and they assume
that by speaking of ontological neutral
neutral nature of reality they are
trying to build open uh the mathematical
nature of markovian blankets markovian
models
to make concession on some sort of
monism which identifies the nature of
reality as neither physical nor mental
uh but i would say that it would be
quite a lot to read into
the use of mathematical models so
of course mathematical tools are neither
in some sense
are not neither a mental nor physical if
you are not

an idealist or realist especially if you
are not a physicalist or idealistic
mathematical entities but it doesn't
follow that the target systems that
could be used
on the basis of using these mathematical
tools itself
the reality the target system that is
going to be used by invoking this sort
of mathematical tools itself is neither
mental nor physical so this is the
conclusion of the paper and for
coming to that conclusion i have also
discussed some sort of
structural reality interpretation of the
paper again it is
about the relation between mathematical
tools and the physical reality that they
represent
so uh again fristen and colleagues would
be happy to
uh make conversation on some sort of
structural realities
uh interpretation of their markovian
modernism
but in the context of the structural
realism too a structural realist as
some of the leading structural realists
such as steven french and
james lady man are quite explicit about
this clear
they do not make their ontological
commitment on the basis
of the mathematical framework that they
use so
they use mathematical framework they use
modal theory they use set theory because
they believe that

this semantic framework modal
theoretical approach or
set theory capital to science uh virus
it's
uh structuralist loyalties on its
sleeves but it doesn't mean that
ontology itself could be read off from
the features of this
mathematical tools that they use so
they are very clear that the job of the
mathematical tools that they use so
uh orthodox structurings are using
mathematical tools such as set theory
and modal theory and category theory
but they are very clear that the job of
these mathematical tools is mainly
representational it is not
constitutional it is not ontological on
the other hand it is the physical
reality that is supposed to be
represented
that has some sort of ontological
constitutional rule
and with regard to markovian monism if
you want to derive any sort
so the use of the mere application of
markovian
doesn't lead to any sort of metaphysical
conclusion
because markovian models are themselves
mainly representational tools they
help us to represent features of the
reality
that is the target system the brain or
consciousness
or organism or the environment or
whatever attitude
it doesn't itself we could not build

upon the features of this mathematical
model

itself to derive any source of
ontological or metaphysical contribution
of the features of the target system

so uh thank you very much for listening
and uh i

i hope that what i said makes some sense
at least to some of you

but although i could not be completely
sure so thank you very much

awesome thank you for that fascinating
presentation

we already have a couple questions in
the chat so if anyone else wants to
ask questions we'll have a good amount
of time now just to

talk a little bit and i'll just read um

down the questions that were asked in
the chat and a few things i wrote down
so the first question in the chat was um
as far as different styles of monism or
different

threads in the history and the
philosophy of monism

how does a dawkins based monism
so thinking about like memes biological
or non-biological mimetics

that's kind of a dawkins monism how does
that parallel or overlap with fristonian
monism so in other words how does
friston's monism

or the markovian monism relate
or overlap with other approaches to
monist

type theories

uh well uh thank you very much

uh for the question i'm so i'm not very

knowledgeable about the different types
of monies and
so i knew about some quite classical
theories of
modernism i know a few things about for
example
are russellian monism or uh the form of
monism that has been developed by
char sanders pairs aside from for
example the classical forms that are
developed by espinosa
but so i'm not very well versed
i'm not very knowledgeable about this
kind of dark kinsey and momentism
i barely know anything about memes
and i'm not sure that we could identify
men as real ontological
entities but uh and i have already
argued that i have some sort of
uh misgivings about
uh pristine and these uh and hobson's
version of monism so
uh i do not want to give a ready-made
answer
i would say that if the arkansas monism
are resembles
a fristonian monism with regard to the
fact that it
involves memes which are uh
as far as i can guess some sort of
theoretical tools
as features of reality uh it could be
liable to the same kind of objections
that i have
uh experienced here against
the fristonian kind of momentum
thanks for the clear answer and i guess
the follow-up on that is

where do you situate this markovian
monism

in relationship to these classical kinds
of monism that you just mentioned is
that discussed in the paper or what

how would you frame that so

uh all right uh thank you very much
again it is

it is a fascinating question so before
this i have developed some sort of a
structural realist

uh theories of uh consciousness too
that happened basically based so my own
approach

to a free energy principle is associated
with some sort of a structural realism
too

so again i myself have invoked some sort
of mathematical structure

embodied informational structures uh
to describe the structure of the mind
and consciousness and my basic
uh argumentation there was that this
structure

sort of underpin the diversity between
different icons of consciousness in
terms of free energy principle uh gay
org north hopes resting estate activity
theory and for example uh integrated
information theory of consciousness so
i myself have developed some sort of a
structural realist but with regard to
the

metaphysical aspects with regard to so i
see monism

specifically as some sort of
metaphysical approach and

also i have been very cautious with

regard to stating
a metaphysical stance so the version of
a structural realism that i have
defended was some sort of epistemic
structure realism it wasn't still
realism
but it was epistemic in the sense that
uh
i was ready to make commitment to the
fact that to the extent that our
scientific or scientific knowledge
is informing us the future of reality is
a structure but we couldn't make
uh assertions about the nature of
reality itself so
whether it is a structure or not or
whether it is uh
the same stuff uh this the cell same
stuff that is there or there are several
kind of substances so uh i
i profess agnosticism uh about the
features of reality in itself
so the version of a structural realism
that i defended was episodic and i
have no stance on the nature of reality
in itself whether of
monies or dualist types or any other
kind of approach that's really
interesting um
seems pretty fair to be agnostic on the
ultimate structure of reality and you
had a quote on one of the slides that
the structural realism is taking the
pressure off of
strong ontological commitments so
how would you summarize this kind of
best of both worlds
structural realism approach are we just

is it a buffet are we just taking what
we like
or what are we committing to with um
epistemic
structural realism very very very good
so uh i think that it is partly
an issue of motivations so uh when
stating
a structural realism for example people
such as
john laurel uh built open arguments from
history of science to argue that so
uh it takes the pressure of
untold pressure in the sense that it
doesn't
make commitments to the intrinsic nature
of entities it doesn't make
commitment to objects it is committed
mainly to the relations
so uh and i think that uh i'm defending
some sort of epistemic form of a
structural realism because as i say it
is completely agnostic about the
features of reality itself so
for example uh people such as john
warrell are
are defending structuralism on the basis
of history of sciences with very clear
examples for example from
the history of optics to show that there
is some sort of continuity between
theories of optics although it seems
that there are radical theoretical
shifts in the history of science so
some of the terms some of the
theoretical entities that were supposed
to be the subject of anthological
commitments before

are eliminated from our ontology today
but it seems that there is some sort of
continuity at the level of the structure
of the theories at the level of form
so they are not making come and in this
sense they are indeed taking
ontological pressure from uh their
theory however there are
other forms of for example
anti-construction realism
uh which itself comes in in various
strides for example
James Layman is defending some sort of
uh eliminativist
uh approach the structural realism i
think that for good reasons from the
his from recent many particle physics
uh but uh i'm
slightly skeptical that so that sort of
structuralism is taking any sort of
in terms of ontological commitments i'm
skeptical that it is
less uh cumbersome than the
orthodox forms of uh scientific realism
i'm sure that
there is a perfect match between elite
eliminating
structural realism and
what we know about what modern physics
many particle physics is representing
about the features of the world
but with regard to metaphysics for
example antirealist structuralism claims that
all that there is is a structure
which is a strong metaphysical claim so
it
eliminates entities objects from the
ontology which is quite ambitious and a

strong ontological commitment so
and the thing is that i do not think
that the version of
structural realism that freestone and
colleagues have defended is epistemic
because they are defending some sort of
metaphysical thesis here monism is some
sort of metaphysical thesis so
although structural realism was supposed
to be
ontologically modest in the sense that
at least for example world's version
doesn't make commitment
to the nature of
the world itself the metaphysical
versions of a structural realism such as
james lady
are including are harboring strong
metaphysical tests
about the features of reality that they
are claiming to be completely structural
and i think that in this sense markovian
monism that is defended by frieza
and others is not metaphysically modest
at all
and i think that this is one reason so
i didn't and thank you very much for
bringing this point to my attention i
didn't expel
this point clearly in the paper because
i didn't
explain many times to discuss the
structuralism this is specific paper
but i think that not as i look at it
from uh
here i can see that it could be the main
motive
behind my mind that encouraged me to

be a bit critical about this form of uh
monism that is
uh presented in this paper wonderful
paper by fristen and colleagues
cool cool response here's another
question from the chat kind of keeping
with this theme of physics and
realism and scientific communities that
are evolving their models
you mentioned optics so let's think
about particle physics
cambridge breaths wrote can the author
describe
in his personal belief the main
difference between a mathematical
markov blanketness of an electron when
it's behaving as a particle
versus when it's behaving as a wave
so how do we think about quantum
mechanics and
quantum phenomena when we're talking
about these different
realism ideas
well to be quite honest uh i have no
idea
do you think that it is mainly because
of my ignorance of uh
modern physics so i have some general
conception of what happens
at the level of many particle physics
but
how to state that in terms of marco and
i can
imagine how could we state that in terms
of markovian blankets
but i cannot see how the distinction
between a particle being in a vague
state or being

an entity being in a very state or
participant

uh could be related to this issue of
markovian models i assume that markovian
models

so one thing as i know about markovian
models is that they are

so powerful they are such powerful
modeling tools that they could be used
almost to model everything that there is
uh

to be modeled uh but

well and i uh suspect and it is only
some sort of uh wild guess that we could
model the behavior of an
entity either in its state or
in

as far as we try to provide some sort of
probabilistic representation
of the behavior of the entity to model
the conditional independence between
different states of identity

we could involve markovian blankets
marco blankets to do so but

i think that at the same time i miss the
main part of the

uh the core of the question

i think um just maybe guessing from
the context the electron is something
that we might say

we might want to say it ontologically
exists or some

might think it ontologically exists yet
it can display

this sort of bimodal behavior when it
can depending on the double slit and the
context of the observer

it can be acting yeah great so

all right so how does a markovian
blanket does
that so uh uh as i say this
this kind of approach to the behavior
of entities in many particle physics was
what
motivated people such as steven french
and james ladyman to develop forms of
structural realism so it is not only
an issue of an electron the thing is
that when we look at this
uh the behavior of entities at the level
of at the fundamental level
if i don't want to use a fundamental at
the level of many particle physics
they should do all behavior there is
some sort of package that includes
individual objects
and it is the interpretation for example
of the
electron that allows us to identify that
as some sort of individual objects and
there is a package that
doesn't allow us to model it as an
individual object so
there are both so we have to conceive of
that as
both individual individual objects and
non-individual objects and
this seems to wreak havoc with your
ontology of modern physics and
structural realism provides an answer to
that saying that
all right let's consider the common
reality let's
consider the common structures that
underpin
both formulation and the behavior of

electrons for example are stated in
terms of
several sort of statistics that are
developed
from maxwell's war for example both
einstein statistics
and other sort of statistics so i assume
that we could also
invoke some sort of a mark of blankets
to behavior
to to model the behavior of this kind of
entities at the level of subparticle
physics
but uh does the question mean to
indicate that there
would be some sort of ontological gain
to do so
uh the thing is that even in philosophy
of physics the
the discussion of a structural realism
whether we could
invoke some sort of representational
mathematical tools
or we have to just rely on the
geometrical or physical statement of the
situation
so um i think that these are
sophisticated
issues in the uh philosophy of physics
and
and i'm hardly an expert in that field
but
uh i'm not very clear but uh
obvious advantage with a state of
regimentation of a state of subparticles
in terms of markov
blankets would do to the extent that
metaphysically called

stencils or attributes are a christian
or attitude
cool so here are a few um general
questions what does it mean to be
ontologically
neutral so that was brought up and i was
just wondering
what what is the alternative or how
could it be charged or not neutral what
is
that there are several ways of reminding
ontologically neutral so for example uh
a way that i prefer
is to as i say process ontological
agnosticism to
not make any commitments about the
nature of reality itself
but i do not think that it is this sort
of ontological neutrality that is at
issue in freestone and colleagues paper
i think that they want to be
ontologically neutral in the sense that
so this is this traditional
uh division in the philosophy of mind
between physicalism
between dualism for example physicalism
people who think that
the nature of mind is physical and
people who think that
uh mind or at least mental processes are
different from
physical properties so i think that
markovian monism wants to be
ontologically neutral in the sense that
it doesn't
want to make commitment to either
mentalism or physicalism
it doesn't want to identify

consciousness

uh either in physical terms or mental

terms and in this sense it

strives to remind ontologically neutral

which i think it is quite cool

so as i say i'm a great admirer of this

project because i think that it is the

first significant attempt

that tries to develop free energy

principle into

a remarkable metaphysical testis and i

think that if

free energy principle theories could

really show that the theory

could support or could be construed

along the lines of

neutral harmonism to show that it really

supports a stance that doesn't commit us

either to conceiving

of the nature of mind as a mental state

or

as a mentalist or mental properties or

as physical

it will be a real achievement i think uh

but at the same time i think that just

depending on the mathematical uh

framework of markovian models doesn't

provide

sufficient purchase for doing that it

doesn't provide

a sufficient basis for boosting a

well-supported form of monism

very cool so it sounds like there's the

eternal debate of whether mind leads to

matter or matter leads to mind is there

one type of thing

two kinds of things and it would be

awesome for

free energy principle to step into that gap

but what we've seen so far with the interpretation

of the mathematical construct of the markov blanket does not adequately resolve this metaphysical question so what kind of

analyses or research or measurement or calculation i mean i'm a scientist so i'm thinking about measurement and experiments but as a philosopher what would um reduce our uncertainty about where the fep fits into all of this so uh far far it be from a philosopher to try to prescribe or endorse anything to

the scientists the thing is that i feel that so

so a cool thing about cognitive sciences and

neurosciences is that i think that uh philosophy car

ideas could really be useful to scientists in this field but

uh the nature of this collaboration the nature of what happens here is so sophisticated that

one can hardly so so i'm told that prediction is

pretty hard especially about the future so

i i couldn't have any idea or any prescription

what can we do to bring to to

make this approach more fruitful the only thing that we could i think

we could do is precisely the thing that

we are doing
just trying our best and uh
collaborating so
and i think that this is why this active
inference lab is a cool idea because
it allows people from different
background come
and share their ideas and we will see
whether anything
uh significant will emerge from all of
that or not but
i think that this is the only thing that
we could do oh it has been the only
thing that we have been doing for quite
a while and paid well those are
cool very it's almost like there's the
mind matter dualism and then there's the
science and the humanities or science
and philosophy and so
is one going to envelop the other is
there really only one
and the other one is kind of a sub field
and instead of
playing into that dichotomized debate
which is polarized it's almost like
we're taking an ontologically neutral
stance here and highlighting the
importance of
coming together across different
contexts and having that
conversation so my question on that
would be let's just say that there was
somebody who were curious
and wanted to jump in to learn more
what kind of research or skills or
mentorship
might help this person on their journey
whether they're

you know from a philosophical side or
from a scientific side
what do you think would be some key tips
for somebody who's just starting on
their
fep journey uh
well um the best thing that i've heard
about the relationship between these
different disciplines comes from
a stephen french who is a structural
realist and also a wise philosopher of
science saying that
metal physics is best when informed with
science and science is best then
uh i don't want to say that informed by
metaphysics but that is
taking some metaphysical insight into
consideration so
but on the other hand i think that the
nature of
uh scientific and academic work these
days
is so so in other words the scientific
activity is so
compartmentalized these days people are
doing
their different works in different
disciplines in their own labs
and uh the nature of the problem is so
hard that
they have to spend all of their time on
specific problems so uh
i can hardly also provide any advice of
what kind of mentorship suits anyone who
wants to
work hard up do some excellent job in
frener's principle what i can say that
what's very interesting about free

energy principle is that it seems to provide at least some uh purchase for for a discourse some space for a discourse between scientists and philosophers so it seems that some of us some of the uh good scientists such as for example maxwell ramstad or uh alex constant or fristen himself who are scientists are are completely interested in philosophical discussions of what's happening here and it seems that what's even more wonderful it seems that uh philosophical discussions are affecting the nature of their work in some sense i think that it is something that you can hardly observe for example in uh age wr theories of physics these days uh i don't know whether the great physicists are studying philosophy for their amusement or not but i doubt that they are in this sort of constructive uh relationship with uh philosophers so uh i think that this kind of and the nature of free energy principle itself is quite unificatory brings different disciplines because uh different ideas from different disciplines together so i have no clear idea about the mentorship or any kind of schedule or program but i can say that people who are interested in free energy principle and

studying that
have already chanced to get a wider
scope
with wide intellectual scope with
insights from a number of different
disciplines biology computational
neuroscience
uh cognitive psychology philosophy of
mind
uh philosophy of biology and that sort
of thing so it seems to come with the
discipline as it is
i

[Music]

and i liked how you grounded the paper
and the presentation
in the history and the philosophy of
science because
it is a community that enacts the
conversation
around the ideas and then when there is
not a healthy dialogue
it's kind of like the philosophers are
uh sweeping up after the scientists or
vice versa or they're just sort of
spectators to each other
but then we're almost moving towards a
a fruitful bi-directional conversation
that actually as you're pointing out
with the specific people who are
taking this into their research approach
it actually is something
where the science and the philosophy are
co-influencing each other
so that's very interesting and special
it's almost like it's such a big
scientific idea and so challenging that
we know that we need multiple expertises

we know that we're gonna need to be
working on it together for a long time
and
then that helps set the stage
for something that might surprise people
because it's not just coming from their
camp so we have history and philosophy
science and then i guess another thread
is math
mathematics markov blankets as you
pointed out came from
pearl in the 1988 work and has been
developed
a lot subsequently where does set or
category theory
come into this uh
well well
uh very good question
i think that in some sense it depends on
your view
of your your perspective on mathematics
so
if you are set or taking uh foundations
in mathematics
if you are a mathematician who uh
prays to the god of set theory every
night in his loneliness you naturally
think that every kind of mathematical
theory
is coming is based on sectarian security
has a general foundation or
there are other people who are ready to
give this
rule to the category theory but
uh for me because my uh engagement with
these ideas come from a background of
structural realism
the main thing is that uh

and uh it has been going to
the wonderful works of uh
soups uh patrick soups and van frozen
and
philosophers of science came to so so
rational reconstruction have been a
program from the very beginning of the
history of philosophy of science
so philosophers and again it is why free
energy principle
is so interesting for me because because
of its great unifi
unificatory scope and there have been
philosophers of science and there are
still philosophers of science who think
that the main job of philosophy of
science is to
form some ideas about unification of
sciences and in order to bring about
this unification
uh philosophers thought of using
logical formal frameworks for state
regimenting different theories from
different disciplines so
for example in physics people are using
a special form of mathematics
in biology when they are using
mathematics at all they may use a
statistical
uh theories or different forms of
mathematics
and they came to this idea if that if we
use some form of logic some form of for
example set theory model theory
we could have a cool mathematical
framework that allows us to regiment
different theories from different
disciplines

and clarify their relationship together
and have a
realistic clear
neatly stated view of what's happening
in the world assuming that science
provides a reliable picture of the world
if we
could use model theory or set theory
and the thing is that uh and i stated
this view
in a book that has been published by
spring you're according to a structural
realism
i begin to have doubts about the
usefulness or at least the realistic
impact of using set theory and model
theory to represent the features of the
world so
i become a skeptical in the sense that i
thought that this
uh set theory and model theory uh
frameworks are too abstract and even if
we could regiment
the structure of sciences into these
frameworks
we still couldn't account for scientific
representation because these are
the set theory and category theory do
not represent the word these are
scientists who are representing the word
using mathematical models
so i try to
depart from the usual ways of scientific
representation on the basis of sectarian
model theory
and invoke other sorts of mathematical
frameworks that could
uh provide a better match a more

congenial tool for accounting for scientific representation it was very beginning to uh try to introduce some sort of information theory framework and not only information to article anymore but embodied information sort of framework meaning precisely the same sort of framework that fristen and colleagues are using in terms of approximate uh bayesianism to regiment the structure of scientific theories because i came to the idea that it allows us to have a more realistic account if we be able to regiment the structure of theories into this approximate bayesian framework for example instead of say theory and modal theory we could provide a better understanding of the relationship between scientific theories which are actually embedded in the brain of scientists or at least in the scientific community and their target system in the world because uh uh their set theory and model theory are too abstract to take care of the job uh this kind of approximate bayesianism is something that at least according to our best scientific theories is by default embodied in the brain of organism and in this sense we could uh have scientific representation super window the relationship between the organism

and its environment but it is another story it is not directly related to this markovianism thing that i developed here well it is it's super interesting because this embodied and enacted relational approach we can't leave the scientist out of the picture so it's almost like we we remove too much when we abstract that relationship away and then it almost hints towards that electron question from earlier which is well if your ontology or your model is just an electron not even the context you're going to be lost because you won't understand why certain things are happening to your system of interest and so how do we pull back rigorously and include enough context setting so that there is an ontology or a local logic that's making sense so it's just cool like how the historical thread the philosophical thread it's not um spice or salt on the research like just adding it up it's actually something that is is um intertwined and and sets the total scope of the scientific journey so it's like very cool to hear about that so if there's any other questions in live chat we'll ask but

yeah only with regard to a quick follow-up about the last point that you mentioned so part of that might be the contemporary philosophy of science at least the dominant approach which is structural realism is that precisely as you mentioned it leaves those human beings out of the picture when it comes to scientific representation so uh there were a bunch of brilliant philosophers of science such as ronald geary and for example nancy narcissian who is fortunately still active who try to provide some sort of cognitive model of scientific practice itself of scientific activity but for some reasons i i'm not sure about the reason it has been pushed aside uh and it is not the dominant view of philosophy of science again so part of my activity in the philosophy of science was to bring this sort of cognitive approach back to the foreground and i think that i was fortunate that i bumped up to this free energy principle and predictive processing because in order to account for cognitive models of science so we uh sometimes we do not just want to sit in your armchair and do philosophy of science but want to support our philosophy of science by some actual scientific theories and

in order to develop these
cognitive models of science cognitive
approach to philosophy of science
it seems that free energy principle and
predictive coding
are fascinating tools because they allow
us to provide a very good
well supported picture of the
relationship between
scientists who are basically organism
and the community of scientists
uh which could be uh specified in terms
of
i don't know free energy principle
common theories of communication under
free energy principle
the relationship between community and
scientists on the one hand and
their environment the world on the other
and i think that this picture is much
more realistic than the ordinary
but the orthodox picture of scientific
representation
that is fashionable yep cool it's like a
yes
and with action and inference so
philosophy isn't just the armchair
what does it look like to be embodied
and inactive
and then perhaps it looks like what is
known by the career name of science
but we know that learning and research
happens inside of academia outside of
academia
so it is a more realistic picture and
here's a related question from the chat
stephen asks this ability to dissolve
the internalist and externalist dualist

position

is a powerful part of markovian monism

thus giving an

open dynamical system uh uh

the adaptive open boundaries any

thoughts on this yeah i completely agree

so

in the beginning i told that i'm a great

admirer of markovian monism i think that

it is a bigger step forward and this is

precisely because of this reason

because it goes beyond this kind of

orthodox dualism between

intel so here it is described as some

sort of dualism between internal spaces

and external spaces

and um frison and colleagues put a lot

of force and energy into developing this

kind of informational geometries that

again i think

is a wonderful uh formal framework to

model this idea so

this is precisely uh in my opinion for

what it is

this is precisely the valuable part of

markovian modernism that it

tries to use this kind of scientific

model

that goes beyond this dualism between

internal spaces and external spaces to

develop some source of uh

uh some sort of hormones and

metaphysical tests so

i completely agree that this is a

brilliant idea so

the question is what source of

philosophy called argumentation or

scientific argumentation

could be offered to support this picture
so i think that's as it is the status
quo of this
theory is not strong enough to carry the
metaphysical weight the question is
and as i said earlier the paper uh
aims to come with the hope to provoke
the free energy principle theories to
develop this idea to develop markovian
modernism so what sort of philosophical
or scientific
argumentation could be offered to uh
support this insight that has been
mentioned by
stephen and lies at the core of
markovian mode this is the main question
awesome so just in the last um few
minutes here
this paper came out you know one week
ago you probably wrote it several
months before but what are your next
research questions or what
are some exciting ongoing threads like
where do you go
let's say someone read this paper where
are you taking
this frontier of thought uh
all right again uh oh cool idea so
uh so as we have already gathered
uh my mind is very disorderly and it is
because i'm cursing several lines of
research together so
uh part of my research right now is to
develop this kind of model-based
understanding of markovian monism
to defense of modularity tests it's not
precisely the
source of fedorian modularity but at

least the source of modularity that are defended by for example the alcohol and i think that the model-based approach provide at least some some grounds for defending this source of modularity this is on the other hand i'm trying so uh the other thing is that i feel that uh there is ample room for good metaphysical work on this free energy principle and free energy principle theory of consciousness so uh as you may know there are already so markovian monism for example offer some sort of metaphysical realist reading of uh theory of consciousness but there are also illusionists constitutes illusionist interpretations of a theory of consciousness under free energy principle and i think that for some reasons none of this provides a quiet good metaphysical interpretation of theory of consciousness under free energy principle so i'm developing another theory i'm trying to develop a theory of consciousness under free energy principle that could be construed along the line of uh strong intentionalism and it is associated with uh works of historical works of uh friends barantano and tinkering more recently so

uh it is another line of research that i
have in mind to associate
this theory of consciousness with some
sort of intentionalism
uh and then again uh i'm engaged with uh
as as the paper may indicate i'm looking
for further groans for
for the reasons to develop or at least
support this markovian monism
myself so as i say this paper is just
the first
the negative part of some sort of
argument that has to be produced by
others or myself so
uh i also i'm trying to
because i because i can't see the appeal
of this markovian monism so i try to
find
reasonable groans for supporting this
theory i'm not sure
where does it go but yeah it is
an awesome approach a lot of times a
critique it's almost like
consequentialism someone says i don't
like your outcome
so i'm going to look for all the details
or say that it wasn't really supported
conclusion
and you're almost like saying actually i
do
see that goal or i see the benefits here
and we just need to navigate in a
slightly
new way but we're going to be learning
and connecting these ideas along the way
because it isn't mapped how we're going
to get to markovian monism and maybe
that's not where we're going

but we're working towards brilliant i
just
get excited and caught you because it is
related to another part of my research
which
tries to relate free energy principle to
theories of
group group knowledge and uh to account
for for example
collective intentionality on the basis
of free energy principles so
uh freestone and fears have produced a
very fascinating theories of
communication and social relation under
free energy principle
and i think that i think that
this theory could be applied to develop
the part of the epistemological
social epistemology and i think that
there would be wonderful
wonderful reasons so uh it is natural
for me to
see this kind of scientific back and
forth between different
uh intellectuals as some sort of uh
collaboration that
consequentially consequentialist as you
phrase that masterfully
that hopefully leads to the growth of
knowledge in general
yes the individual in the armchair it
lends ones toward
it lends ones almost to neuro
reductionism and individualism
but when the person is walking around
and at a conference and they're
they're sharing their experience and
they're learning and being surprised by

by friends and colleagues then there
really is something that's more like
distributed cognition so i'll really
look forward
to seeing that um majin thanks so much
for this
appearance this was a epic and special
first guest stream for our lab and
you're
always welcome just let us know if you
have any other
topics that you want to come on a stream
with you or colleagues or students or
anything we're always there to
host uh and hold the space for this kind
of work so
again thanks a lot for the research and
i'm sure everyone really appreciated
hearing it
thank you very much for having me and
also cool work to
to do this uh active inference lab so uh
good luck with that ah and very well
done and thank you very much for
having me great we'll be in touch so see
you later